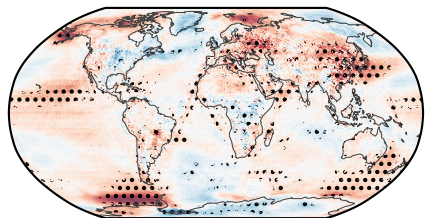


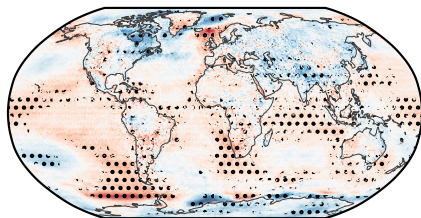
ENSEMBLE-MEAN emulator minus held-out CESM2 (6-11 emulator vs 6-11 CESM2 members), 10-year mean anomaly vs 1850-1900
 hatching = difference significant at $p < 0.05$ (Welch t-test, FDR-controlled at $q = 0.1$); precipitation as % of its 1850-1900 baseline

Historical
2005-2014 (10 yr)



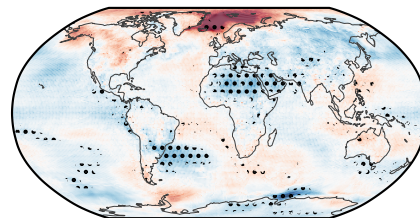
$r = 0.975$ RMSE = 0.159 °C
18% of area significantly different

SSP3-7.0
2091-2100 (10 yr)



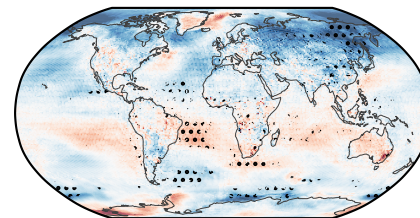
$r = 0.997$ RMSE = 0.146 °C
24% of area significantly different

Aerosol-only (AAER)
2041-2050 (10 yr)

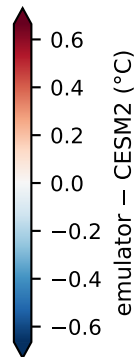


$r = 0.944$ RMSE = 0.145 °C
9% of area significantly different

Greenhouse-gas-only (GHG)
2041-2050 (10 yr)

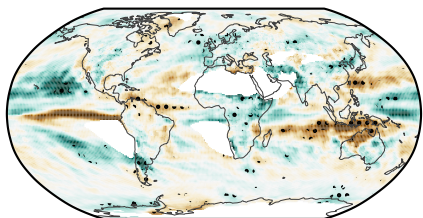


$r = 0.985$ RMSE = 0.202 °C
8% of area significantly different

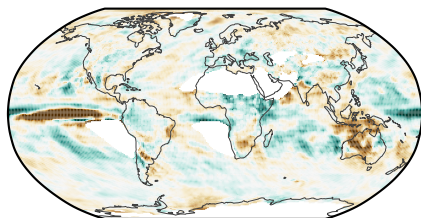


Temperature

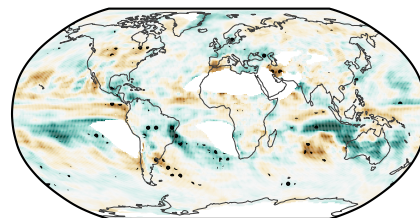
Precipitation



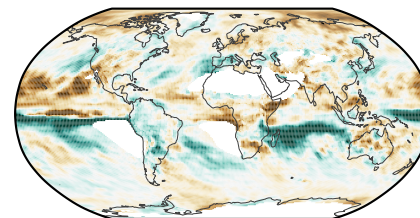
$r = 0.807$ RMSE = 4.751 %
6% of area significantly different
(9% arid, masked)



$r = 0.991$ RMSE = 5.114 %
0% of area significantly different
(9% arid, masked)



$r = 0.930$ RMSE = 3.865 %
3% of area significantly different
(9% arid, masked)



$r = 0.968$ RMSE = 5.875 %
0% of area significantly different
(9% arid, masked)

